1.(i) The drone deviates from its original trajectory π_{dr} between steps $p+1$ and $q-1$. During $[p+1, q-1]$ it first moves to a truck by traversing, say i, edges. And after recharging completes it comes back to π_{dr} by traversing, say j, edges.

That means between step $p+i$ and $q-j$ the drone rides on a truck with zero cost. We have to minimize i and j which are the deviations between the given steps p and q.

Minimize i, j

subject to

$$a_{dr}^{p+i}, \dots a_{dr}^{q-j} \in A_{tr} \quad \left.\right\} \text{Assigning to a truck between } p+i \text{ and } q-j$$

and

$$\left.\begin{aligned} e_{dr}^{p+i} &= e_i^{k} \\ e_{dr}^{p+i+1} &= e_i^{k+1} \\ &\vdots \\ e_{dr}^{q-j} &= e_i^{k+q-j-p-i} \end{aligned}\right\} \text{edges of drone should match with truck } i$$

$$i \in A_{tr} \quad \} \text{ some truck } i$$

$$e_i^{k} \dots e_i^{k+q-j-p-i} \in \pi_i \quad \} \text{ trajectory matching}$$

1.(ii) The strategic form game is as follows:

$$\Gamma = \langle N, (S_i), (u_i) \rangle$$

where $N = \{1, 2, \ldots, n\}$

$S_i = \{1, \ldots, K\}$

$$u_i = \frac{1}{m} \quad \text{if } |s_i - b| \leq |s_j - b| \ \forall j \neq i$$

$$= 0 \quad \text{otherwise.}$$

Here, $b = \frac{2}{3} \cdot \frac{1}{n} \sum s_k$ where $k \in N$

and $m = \left| s_k : |s_k - b| \leq |s_j - b| \ \forall j \neq k \right|$

2.(i)

Given that (C, C) is a strongly dominant strat eq.

This means, C is a strongly dominant strat for player 1

and C " " — " " player 2

For player 1: [according to the def of dom strat eq]

$$u_1(C, NC) > u_1(NC, NC)$$

$\Rightarrow -x > -6 \Rightarrow x < 6$ —— 1

$u_1(C, C) > u_1(NC, C)$

$\Rightarrow -x > -3 \Rightarrow x < 3$ —— 2

For player 2:

$u_2(C, C) > u_2(C, NC)$

$\Rightarrow -x > -3 \Rightarrow x < 3$ —— 3

$u_2(NC, C) > u_2(NC, NC)$

$\Rightarrow -x > -6 \Rightarrow x < 6$ —— 4

Combining 1 – 4, the answer is $x < 3$.

2. (ii) For player 1: [according to the def of weakly, but not strongly dom strat eq.]

$u_1(C, NC) > u_1(NC, NC)$

$\Rightarrow -x > -6 \Rightarrow x < 6$ —— 5

$u_1(C, C) \geq u_1(NC, C)$

$\Rightarrow -x \geq -3 \Rightarrow x \leq 3$

Similarly, if the $>$ and $\geq$ flips, we have

$x \leq 6$ and $x < 3$

For player 2:

$u_2(C, C) > u_2(C, NC)$

$\Rightarrow -x > -3 \Rightarrow x < 3$

$u_2(NC, C) > u_2(NC, NC)$

$\Rightarrow -x \geq -6 \quad \Rightarrow x \leq 6$

Similarly, $x \leq 3$ and $x < 6$

So, now we have the following

$\boxed{x < 6, \; x \leq 3}$ ✓ or $\boxed{x \leq 6, \; x < 3}$ ✗

and

$\boxed{x < 3, \; x \leq 6}$ ✗ or $\boxed{x \leq 3, \; x < 6}$ ✓

$x < 3$ is not possible, because then we have strongly dom strat eq [as shown in 2 (i)]

So, we ✗ these constraints; and the remaining two constraints marked ✓ applies.

Combining ✓ constraints, we have $x = 3$;

Please note, we cannot have $x \lneq 3$ as $x < 3$ gives strongly dom strat eq.

So, the answer is $\underline{x = 3}$.

2. (iii) For either of the above eq., no player has an incentive to unilaterally deviate from its strategy (in this case, C for both the players).

So, (c, c) is a self-enforcing strat profil

3. (i) In the given game, (8, 20) and (20, 0) are the Pareto-optimal outcomes.

This is because, given any of the above outcomes, there is no other outcome s.t. both the players can simultaneously improve (at least one player does not decrease) their payoffs.

3. (ii) Pareto dominated means an outcome (A) is strictly worse than another (B) because at least one player is worse off, while no one is better off.

In the given payoff matrix, there are many pairs of outcomes, for example,

[(6, 6), (10, 0)] ; [(6, 6), (8, 0)]

[(20, 0), (6, 6)] ; [(8, 20), (20, 0)] and so on.

You have to mention only one such pair.

3. (iii)

	X	Y	Z
A	6, 6	8, 20	0, 8
B	$\boxed{10}$, 0	5, 5	2, 8
C	8, 0	$\boxed{20}$, 0	$\boxed{4}$, 4

As we have studied in the class the □ ○ method the □ and ○ coincides in (C, Z).
So, (C, Z) is a PSNE of the given game.

Alternative way: Best respons correspondence.
(though both the methods are exactly same).